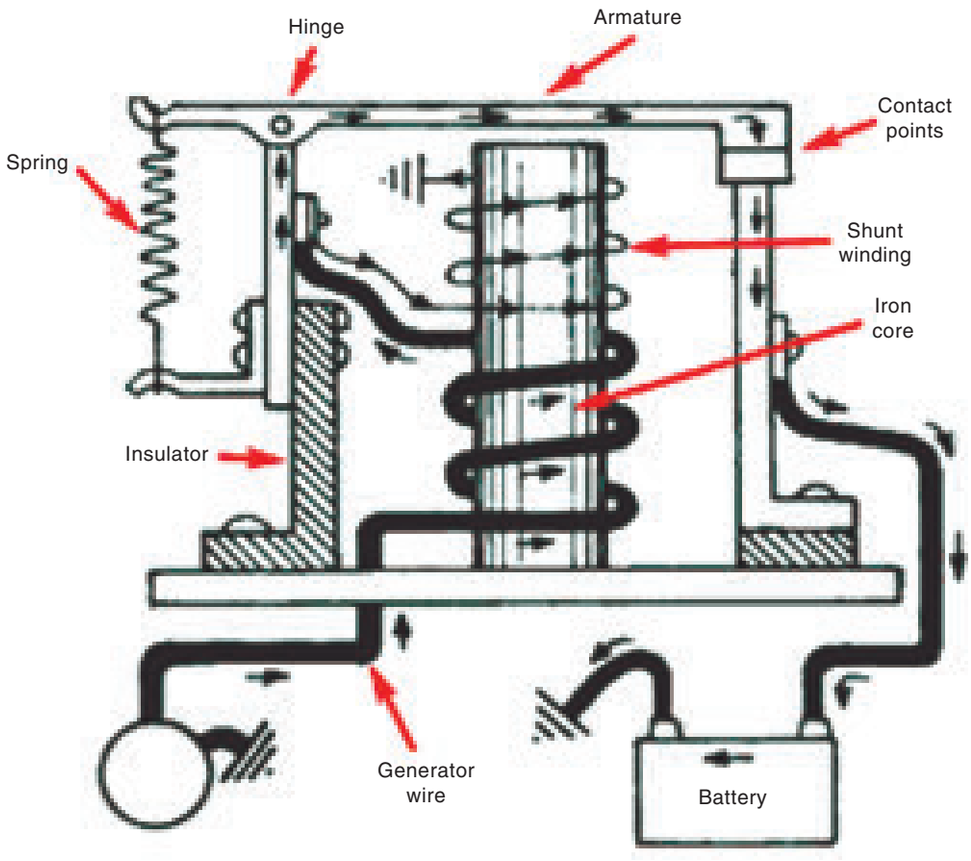


Figure 10.9.2 (a) Cut out relay

10.9.3 Alternator

The alternator converts mechanical energy from the engine into electrical energy. It is also called A.C. Generator. With the help of rectifiers formed by diodes, the Alternating Current (AC) is converted into Direct Current (DC) and stored in vehicle battery.

The rotor shaft usually consists of pole pieces with a field coil or winding between them. When the current flows through the winding, it becomes an electromagnet. Current flows in and out of the winding through two brushes riding on slip rings. Each slip rings connects to one end of the winding. The current produces a magnetic field which rotates as the rotor turned by engine crankshaft through a belt. The strength of the magnetic fields varies with the amount of the current flowing. The stationary loops or conductors are assembled into a laminated iron frame. The assembly is stator. As the rotating magnetic field cuts through the windings, an alternating current is induced in the stator. As the magnetic poles are changing for every half revolution, stator induces an alternating current for every revolution of the rotor.

Advantages

- Light in weight
- A high output is obtained even at low engine speed.
- The cutout relay is not needed.
- Long life.
- Highly reliable.

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